IAN PIKE

C++ SOFTWARE DEVELOPER

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github.com/Rinzii | ianpike.com | LinkedIn

C++ developer focused on numerical libraries, graphics infrastructure, and language design. Open-source maintainer and member of ISO WG21 and INCITS/C++.

TECHNICAL SKILLS

Languages: C++17/20/23, C, Python, C#, SQL

C++: constexpr, templates, floating-point arithmetic, memory management, API design

Tools: CMake, Git, GitHub Actions, GCC, Clang, MSVC, CTest

Graphics: Vulkan, Direct3D 12, Metal

SELECTED PROJECTS & CONTRIBUTIONS

CCMATH | CREATOR AND MAINTAINER

2024-present

github.com/Rinzii/ccmath

- Develop a header-only C++17 math library with a shared API for compile-time evaluation and runtime computation.
- Fixed constexpr remainder rounding and fmod errors with large quotients, adding compile-time and runtime regression tests for subnormals, signed zero, and rounding modes.
- Build numerical tests for floating-point edge cases and rounding behavior, including comparisons against MPFR and CORE-MATH reference implementations.
- Maintain CMake builds and GitHub Actions test workflows across Linux, macOS, and Windows with GCC, Clang, and MSVC.

AZOTH RHI | INDEPENDENT GRAPHICS PROJECT

github.com/WizWorksInc/azoth-rhi

- Develop a C++23 render hardware interface with Vulkan, Direct3D 12, and Metal backends.
- Implement command recording and resource synchronization, with conformance tests for barrier states and resource transitions.

LIBASSERT | OPEN-SOURCE CONTRIBUTOR

github.com/jeremy-rifkin/libassert/pull/93

- Contributed platform and feature-detection infrastructure for optional C++20/23 support while retaining C++17 compatibility. Accepted and merged by the project maintainer.

C++ STANDARDS

MEMBER | ISO WG21 AND INCITS/C++

- Co-author of P1219R3, Homogeneous variadic function parameters, covering typed parameter packs, overload resolution, and compatibility with existing C++ code.

EDUCATION

LAKELAND COMMUNITY COLLEGE | KIRTLAND, OH

Expected 2027

Associate of Applied Business, Computer Science / Software Engineering

PREVIOUS EMPLOYMENT

TERVES INC. | MACHINIST / FOUNDRY WORKER Mar 2021-Nov 2022

STAPLES INC. | TECHNOLOGY ASSOCIATE & TECHNICAL CUSTOMER SUPPORT 2019-2020

AWARDS

SkillsUSA state Computer Programming winner (2019, 2022). Hyland hackathon team winner (2020).